

Displacement response and Loewner growth for the half-BPS Wilson line

Locality, memory, and the limits of an arithmetic reading

Drafted for Edward Baker

24 September 2026 — first draft

Abstract

We study the response of the half-BPS Maldacena–Wilson line of $SU(N)$ $\mathcal{N} = 4$ super-Yang–Mills theory to shape and internal deformations, and the expectation values of a family of contours grown by the Loewner equation, with the aim of deciding whether any of these observables can serve as a causal channel with memory. Four results are established. (i) About the straight line, the quadratic Euclidean shape response is fixed by the Bremsstrahlung function B up to one perimeter constant, and the linear retarded response is local: $\chi_R(\omega) = m_R\omega^2 + 2\pi i B\omega^3$ for every conformal line defect with a dimension-two displacement. The total work reproduces the Correa–Henn–Maldacena–Sever radiated energy, but the renormalized finite-time account carries an indefinite Schott term. (ii) The Loewner trace closed by a return chord with reversed scalar charge (the flipped return) is a regular observable in every regime. Its first two smooth-field evolution equations are exact; in the conformal theory the Loewner time collapses to $s = a\sqrt{t}$; the continuum short-time law is governed by the near-BPS corner of the generalized cusp, whose coefficient is one-loop exact order by order with effective coupling $\lambda|s|$, and which at strong coupling crosses over to $\sqrt{\lambda|s|}$ with an explicit classical coefficient. (iii) A finite-mass source, the string endpoint at $z_m = \sqrt{\lambda}/2\pi m$, is the first channel with memory. Its linearized dynamics are exactly $F = m\ddot{Y}$, $X = Y + z_m\dot{Y}$, $m\ddot{X} = F + z_m\dot{F}$; its energy account is positive at every time; and the boundary field of the moving source is exactly local in the retarded time, with a closed-form quartic transfer function, because the string-depth integrand is a total derivative. (iv) The quartic straight-line response is the first nonlocal one; its memory exponent is set by the non-protected dimension $\Delta_6(\lambda)$, but the straight line has no scale. None of these channels has a power front with a variable exponent or a hierarchy of fixed delays, so no arithmetic comparison is warranted; the physical results stand on their own.

Status. First draft, assembled from three research sessions of 23–24 September 2026. Analytical results are classified statement by statement in the section ledgers; 472 standard-library diagnostics accompany them (Appendix A). Same-assistant audits with separate referee contexts were carried out; no independent specialist review has taken place. Nothing here constructs an arithmetic realization or bears on the Riemann Hypothesis.

Contents

1	Scope, conventions and the question asked	2
2	The straight line: quadratic response, locality and energy	3
3	The Loewner family and the flipped-return loop	5
4	The near-BPS corner of the generalized cusp	8

5	A finite-mass source: memory, energy and an exactly local boundary field	10
6	The quartic straight-line response	12
7	Assessment and outlook	14
A	Numerical evidence, provenance and acknowledgements	15

1 Scope, conventions and the question asked

1.1 The question

The parent program asks whether a physical response can realize the shifted completed-zeta transfer

$$H(p) = \xi(\tfrac{1}{2} + p), \quad K_\omega(p) = \frac{H(p - \omega)}{H(p + \omega)}, \quad (1.1)$$

as a causal, passive input/output map in an ordinary signal norm. Any candidate must first be a physical channel with a defined input, output and energy balance; only then does a comparison make sense, and the comparison proceeds through ordered tests—a parameter dictionary, an identity endpoint, a power front $(2\pi/p)^\omega$ with its logarithmic tangent, the complete local response, and a hierarchy of delays at $\log 2, \log 3, \dots$ with prescribed coefficients—each of which stops the comparison on failure. The theory is not to be adjusted to the target. This paper reports what $\mathcal{N} = 4$ super-Yang–Mills theory, through its half-BPS Wilson line, offers at each stage.

The line defect was chosen because its displacement operator has a protected two-point normalization, the Bremsstrahlung function $B(\lambda, N)$ [1], so that a displacement-driven response is computable at every coupling. The Loewner equation entered because a growing trace is a one-parameter family of shapes that could carry a time hierarchy. Both channels were tested; neither carries the memory the target requires. The paper records the physics found along the way.

1.2 Conventions

The theory is $SU(N)$ $\mathcal{N} = 4$ SYM, $N \geq 2$, $\theta = 0$, with Euclidean action $(1/g^2) \int \text{tr}(\frac{1}{2} F_{\mu\nu} F_{\mu\nu} + D_\mu \Phi_I D_\mu \Phi_I + \dots)$, Hermitian fields, $\text{tr} T^a T^b = \delta^{ab}/2$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]$ and $D_\mu = \partial_\mu - i[A_\mu, \cdot]$. Feynman-gauge propagators are $g^2 \delta^{ab} \delta_{\mu\nu} / (4\pi^2 x^2)$ for the gauge field and the same for each scalar. The Maldacena–Wilson line is

$$W = \frac{1}{N} \text{tr} P \exp \int (iA \cdot \dot{x} + |\dot{x}| n \cdot \Phi), \quad (1.2)$$

with larger parameters to the left. The straight half-BPS line is $x_0(\sigma) = (\sigma, 0, 0, 0)$ with constant $n_0 \in S^5$. Defect correlators $\langle\langle O_1 \cdots O_n \rangle\rangle$ are path-ordered insertions divided by $\langle W \rangle$. We write $\lambda = g^2 N$, $B_1 = \lambda(1 - 1/N^2)/(16\pi^2)$ for the one-loop Bremsstrahlung function, and use the exact inputs of [1]: $C_D = 12B$, $C_\Phi = 2B$, the cusp anomalous dimension $\Gamma_{\text{cusp}} = -B(\varphi^2 - \theta^2)$ at small angles, and the radiated energy $\Delta E = 2\pi B \int \dot{v}^2$. At strong coupling $B \rightarrow \sqrt{\lambda}/4\pi^2$ and $2\pi B \rightarrow \sqrt{\lambda}/2\pi$, Mikhailov’s coefficient [3].

The Loewner traces are generated by the chordal Loewner equation with driver u ; for the linear driver $u = at$ the prefix at time t is $\sqrt{t}$ times the prefix generated by the slope $s = a\sqrt{t}$ up to time one. The base of the trace is the origin and the tip is $q(t)$.

1.3 Classification

Every statement below is classified as a proof, a proof sketch, a labelled numerical computation, an observation, a reading or a literature input, in the ledger at the end of each section. Numerical diagnostics are floating checks of analytic statements, not interval certificates; the standard-library programs and their records are listed in Appendix A. Formulas taken from the literature through automated fetch summaries were validated by limits or by symmetry before use, and this is stated where it applies.

2 The straight line: quadratic response, locality and energy

Deform the straight line by a transverse displacement $\epsilon h(\sigma)$ and the internal direction by a tilt $\epsilon j(\sigma) \perp n_0$, and let $Z[h, j] = \langle W[x_h, n_j] \rangle / \langle W[x_0, n_0] \rangle$. The first variation inserts $\mathbb{D}_i h^i + \Phi_a j^a$ with $\mathbb{D}_i = iF_{i\sigma} + D_i \Phi_{n_0}$; both one-point functions vanish. The Euclidean displacement is Osterwalder–Schrader real, $\Theta \mathbb{D}(\sigma) = \mathbb{D}(-\sigma)$, and Wick rotation makes $F_{i0} + D_i \Phi$ Hermitian; the explicit i is not a failure of Hermiticity. At tree level $\langle iF iF \rangle$ contributes $+4$ and $\langle \partial \Phi \partial \Phi \rangle$ contributes $+2$ in units of $g^2 C_F / (4\pi^2 \sigma^4)$, so $C_D = 12B_1$.

2.1 Quadratic response

Proposition 2.1 (renormalized quadratic response). *Assume that (a) the second variation of the renormalized $\log Z$, away from the diagonal, is given by the defect two-point functions; (b) its extension across the diagonal differs from any other by a finite-order local distribution whose coefficients carry powers of the regulator scale, apart from finite renormalization conditions; (c) Z is invariant under translations, rotations, $SO(5)$ and the reflection described below. Then, for every λ and N ,*

$$\log Z[\epsilon h, \epsilon j] = \epsilon^2 \left\{ \pi B \int \frac{dk}{2\pi} (|k|^3 |\hat{h}(k)|^2 - |k| |\hat{j}(k)|^2) + c_R \int |\dot{h}|^2 d\sigma \right\} + O(\epsilon^4), \quad (2.1)$$

and there is no cubic term.

Proof. The separated kernels are $12B\delta_{ij}\sigma^{-4}$, $2B\delta_{ab}\sigma^{-2}$ and zero for the mixed pair (operators of different dimension have vanishing two-point functions on a conformal line, and $SO(3) \times SO(5)$ has no invariant tensor with one index of each kind). The family $|\sigma|^\lambda$ has poles only at $\lambda = -1, -3, \dots$, so the scale-covariant extensions of $|\sigma|^{-4}$ and $|\sigma|^{-2}$ are unique and free of logarithms, with Fourier transforms $(\pi/6)|k|^3$ and $-\pi|k|$. In the h -sector a δ term is excluded by translation invariance, a δ'' term is the perimeter constant c_R , and higher terms vanish with the regulator; in the j -sector the δ term is excluded by R -invariance and δ'' already vanishes, so that sector is universal. Cubic terms need an odd $SO(5)$ tensor (none) or ϵ_{ijk} with three displacements, excluded by the reflection $x^2 \rightarrow -x^2$ combined with a reflection of one internal direction orthogonal to n_0 , a symmetry inherited from a ten-dimensional rotation. $\square$

The perimeter constant is $c_R = -m_R/2$, with m_R the renormalized heavy-source mass; it vanishes in the natural Maldacena–Wilson scheme, where the one-loop kernel realizes the homogeneous extension exactly. With $c_R = 0$ the form is positive in the transverse direction ($\dot{H}^{3/2}$), negative in the internal direction ($\dot{H}^{1/2}$), and zero on the Zarembo profile $j = \dot{h}$, where the two terms cancel because $|k|^3 |\hat{h}|^2 = |k| |k\hat{h}|^2$. In general the cancellation is $(\pi/12)(C_D - 6C_\Phi)|k|^3$, so it holds if and only if $C_D = 6C_\Phi$: a self-contained check of the CHMS normalizations against Zarembo’s non-renormalization [4, 5, 6]. The first correlators not determined by B are the connected four-point functions of the displacement multiplet, which enter at $O(\epsilon^4)$ (Section 6).

2.2 Locality of the linear retarded response

Take $H(s) = H_0 - h(s)D + (\text{seagull})$ with D the Hermitian Lorentzian displacement and h the transverse displacement of the heavy source; the supplied work is $W(T) = -\int_{-\infty}^T \dot{h} \langle D \rangle ds$.

Proposition 2.2 (locality). *For any unitary conformal line defect, in particular for the Maldacena–Wilson line at every λ and N , the Wightman function is $\langle D(s)D(0) \rangle = C_D/(s - i0)^4$ and its commutator is the contact distribution $-(i\pi C_D/3)\delta'''(s)$. The renormalized susceptibility is the polynomial*

$$\chi_R(\omega) = m_R\omega^2 + i\frac{\pi C_D}{6}\omega^3 = m_R\omega^2 + 2\pi i B\omega^3, \quad \langle D(s) \rangle = -m_R\ddot{h}(s) + 2\pi B\ddot{\ddot{h}}(s). \quad (2.2)$$

Proof. One-dimensional conformal symmetry fixes the Euclidean two-point function of the dimension-two primary; its continuation $\sigma = is + 0$ gives the Wightman function, and $(x \mp i0)^{-n} = x^{-n} \pm i\pi(-1)^{n-1}\delta^{(n-1)}(x)/(n-1)!$ gives the commutator. The spectral density fixes $\text{Im } \chi_R = (\pi C_D/6)\omega^3$ and excludes an $i\omega$ term; the even part is $c_0 + c_2\omega^2$ with $c_0 = 0$ because a static displacement is a symmetry; an ω^4 term carries a negative power of the cutoff. Finally $m_R = c_2 = -2c_R$. $\square$

The mechanism is one-dimensional conformal symmetry plus the integer dimension of the displacement: the discontinuity of $(s - i0)^{-2\Delta}$ is a contact term exactly when 2Δ is an integer, and the displacement of a line defect has $\Delta = 2$ in every bulk dimension. Non-contact commutators, i.e. memory, first appear in higher-point functions. The $2\pi B\ddot{\ddot{h}}$ term is the $\mathcal{N} = 4$ Abraham–Lorentz force; at strong coupling its coefficient is $\sqrt{\lambda}/2\pi$ [3]. The internal tilt channel is an exact resistor, $\langle \Phi_a \rangle = -2\pi B j_a$, finite-time passive with no counterterm.

2.3 Energy account

Proposition 2.3 (work). *For a smooth compactly supported drive, $W(\infty) = 2\pi B \int \dot{h}^2 ds$, independent of m_R and of the scheme; this is the CHMS radiated energy including the coefficient. With the cutoff measure $d\mu_\Lambda = (C_D/6)\Omega^3 e^{-\Omega/\Lambda} d\Omega$ and the translation seagull, the impedance is a Foster function, $W_\Lambda(T) \geq 0$ for every T , and as $\Lambda \rightarrow \infty$*

$$W_\Lambda(T) = \frac{\delta m_\Lambda}{2} \dot{h}(T)^2 + 2\pi B \left[\int_{-\infty}^T \ddot{h}^2 ds - \dot{h}(T)\ddot{h}(T) \right] + o(1), \quad \delta m_\Lambda = 4B\Lambda. \quad (2.3)$$

Proposition 2.4 (the renormalized finite-time account is never passive). *Let $m_R \geq 0$ be finite and let h vanish for $s < s_0$ with a finite-order onset of order $n \geq 3$ at s_0 . Then $W_R(T) = \frac{m_R}{2} \dot{h}(T)^2 + 2\pi B [\int_{s_0}^T \ddot{h}^2 - \dot{h}(T)\ddot{h}(T)] < 0$ for all $T > s_0$ sufficiently close to s_0 .*

The proof is the expansion $h \approx c(T - s_0)^n$: the bracket is negative and of lower order than the inertial term. For flat onsets e^{-1/x^α} the ratio $\dot{h}\ddot{h}/\int \ddot{h}^2$ tends to 2 and negativity persists; oscillating onsets give counterexamples. This is the Abraham–Lorentz–Dirac–Schott account: positivity at finite time is carried by regulator-scale inertia that renormalization removes, and the natural scheme $m_R = 0$ is the extreme case. Section 5 shows how a finite physical mass resolves this.

2.4 Consequence

At linear order about the straight line, any channel built from D or Φ_a has a Laplace transfer that is a polynomial: with $\omega = ip$, $\chi(p) = -m_R p^2 + 2\pi B p^3$ and $\chi^\Phi(p) = -2\pi B p$. Changing λ or N changes only $2\pi B$. There is no front $(2\pi/p)^\omega$, no variable exponent, no delayed singularity and no identity endpoint as a parameter varies. The linear straight-line channel is closed as an arithmetic candidate, for every conformal line defect. The closure is scoped to linear order on the straight line.

Ledger

Statement	Status
Quadratic response, Proposition 2.1	Proof under (a)–(c); cubic vanishing a proof sketch
Zarembo cancellation iff $C_D = 6C_\Phi$	Proof
Locality, Proposition 2.2	Proof
Total work equals CHMS ΔE	Proof (not independent evidence for CHMS)
Cutoff passivity and the Schott limit	Proof
Non-passivity of the renormalized account	Proof; flat onsets an observation
One-loop kernel checks	Labelled numerical computation (72 cases)

3 The Loewner family and the flipped-return loop

3.1 Scaling collapse

Proposition 3.1 (conformal collapse). *For the linear driver $u(t) = at$ the trace satisfies $q_a(Tr) = \sqrt{T} q_{a\sqrt{T}}(r)$. In $\mathcal{N} = 4$ SYM, dilatation invariance of renormalized loops, up to cusp anomalous dimensions, gives*

$$W(a, t) = F(a\sqrt{t}, \delta/\sqrt{t}) (\mu\sqrt{t})^{-\Gamma_{\text{tot}}(a\sqrt{t})}, \quad (3.1)$$

where δ is a rounding scale and Γ_{tot} the sum of the cusp anomalous dimensions.

The Loewner time is therefore not an independent clock in the conformal theory: renormalized loops depend on t only through the slope $s = a\sqrt{t}$, and the short-time limit is the small-driver limit. Pure Yang–Mills at fixed flow resolution differs, because there t/τ is meaningful.

3.2 Choice of completion

The trace must be closed to form a loop. Three completions were examined in units where $t = 1$, where the trace and a straight return chord bound a sliver of width $w(y) = (s/6)y(2 - y)$, $y \in [0, 2]$, with cusp openings $s/3$ at base and tip.

Constant scalar on both legs. The opposite legs are nearly antiparallel with the same scalar charge, so gauge and scalar exchange add. At one loop $\log W \approx (3\lambda(1 - 1/N^2)/2\pi s) \log(1/\hat{\varepsilon})$, which diverges as $s \rightarrow 0$; to all orders the near-antiparallel cusp gives $\log W \approx (6c/s) \log(1/\hat{\varepsilon})$ with c the antiparallel-lines coefficient [7]. The $a = 0$ member is not a normalizable control. At fixed flow resolution the loop measures perimeter growth, $W_\tau = 1 + 8t \langle N^{-1} \text{tr} \Phi_{n_0}^2 \rangle_\tau + \dots$.

Tangent-coupled scalar. With $n = MT$ the loop is a Zarembo loop and $\langle W \rangle = 1$ for every curve [4, 5, 6]; the corners are BPS cusps. This is a Ward-identity control, reproduced by Proposition 2.1 at $O(\epsilon^2)$ exactly because $C_D = 6C_\Phi$.

Flipped return. Keep n_0 on the trace and $-n_0$ on the chord. The chord traversed from tip to base with connection $iA \cdot dx - |dx|\Phi$ is the inverse of the base-to-tip transport R_t with connection $iA \cdot dx + |dx|\Phi$, so

$$Q_t = R_t(1)^{-1}U_t, \quad W(t) = \langle N^{-1} \text{tr} Q_t \rangle, \quad (3.2)$$

and at $a = 0$ the transport is exactly the identity. The opposite legs are nearly antiparallel with opposite scalar charge (the no-force configuration), and both cusps have $(\varphi, \theta) \approx (\pi - s/3, \pi)$, i.e. they are nearly BPS. W is even in a . This is the recommended family.

3.3 Exact evolution equations

Write $k = q_1 \dot{q}_2 - q_2 \dot{q}_1$, $\hat{n} = (-q_2, q_1)/|q|$, $\rho = |q|$, $\gamma = |\dot{q}| - \dot{\rho} \geq 0$; hats denote conjugation by $R_t(r)$; $J(r) = \int_0^r r' \hat{F}_{12} dr'$, $\mathcal{G}(r) = \int_0^r r' (\hat{n} \cdot \widehat{D\Phi}) dr'$, $J = J(1)$, $\mathcal{G} = \mathcal{G}(1)$.

Proposition 3.2 (first equation). *For smooth fields and every configuration, $\dot{Q}_t = \mathcal{X}_t Q_t$ with $\mathcal{X}_t = ik(J + i\mathcal{G}) + \gamma \hat{\Phi}(1)$. Since $\int_0^t \gamma = L_{\text{trace}}(t) - |q(t)|$, the scalar tip coefficient is exactly the growth rate of the perimeter mismatch between the two legs.*

Proposition 3.3 (transported derivative and second equation). *For a local adjoint field X on the chord, $\partial_t \hat{X}(r) = r(\dot{q} \cdot \widehat{DX})(r) + [\hat{X}(r), \mathcal{K}(r)] + (\partial_t \hat{X})_{\text{explicit}}$ with $\mathcal{K}(r) = -ik(J(r) + i\mathcal{G}(r)) + \dot{\rho} r \hat{\Phi}(r)$. Hence $\ddot{W} = \langle N^{-1} \text{tr}((\dot{\mathcal{X}} + \mathcal{X}^2)Q) \rangle$ with*

$$\dot{\mathcal{X}} = ik(J + i\mathcal{G}) + \gamma \hat{\Phi}(1) + ik(\dot{J} + i\dot{\mathcal{G}}) + \gamma[(\dot{q} \cdot \widehat{D\Phi})(1) + [\hat{\Phi}(1), \mathcal{K}(1)]], \quad (3.3)$$

$$\dot{J} = \int_0^1 r(r(\dot{q} \cdot \widehat{DF}_{12}) + [\hat{F}_{12}, \mathcal{K}(r)]) dr, \quad (3.4)$$

$$\dot{\mathcal{G}} = \int_0^1 r \left(r(\dot{q}^\mu \hat{n}^\nu \widehat{D_\mu D_\nu \Phi}) - \frac{k}{|q|^2} (\hat{q} \cdot \widehat{D\Phi}) + [(\hat{n} \cdot \widehat{D\Phi}), \mathcal{K}(r)] \right) dr. \quad (3.5)$$

Both equations hold for any driver with a C^2 trace.

The proofs use the ordered-transport variation on the chord and one radial integration by parts, which is possible because the chord connection commutes with Φ along the chord; the commutator terms arise because it does not commute with F or $D\Phi$. Both equations were verified by finite differences on a noncommuting $SU(2)$ background (relative errors 10^{-9} and 8×10^{-7}). The terms $ik[\hat{F}, J(r)]$ and $(ikJ)^2$ combine into $-2k^2 \mathcal{P}$ with $\mathcal{P}$ the ordered $\hat{F}\hat{F}$ moment, exactly as in pure Yang–Mills; the new terms are ordered pairs mixing $\hat{F}$, $(\hat{n} \cdot \widehat{D\Phi})$ and $\hat{\Phi}$ and chord–tip pairs, which do not combine into a single moment.

3.4 Short time at fixed resolution

Configuration by configuration, with $\mathcal{A} = -(2a/9)t^{3/2}$ the oriented area and $L_{\text{trace}} - L_{\text{chord}} = a^2 t^{3/2}/27$,

$$N^{-1} \text{tr} Q_t - 1 = \frac{1}{2N} \text{tr} \left(\mathcal{A}(iF_{12} + D_1 \Phi)(0) + \frac{a^2 t^{3/2}}{27} \Phi(0) \right)^2 + O(t^{7/2}), \quad (3.6)$$

so that in an $SO(6)$ - and parity-invariant flowed state

$$W_\tau = 1 + \frac{2a^2 t^3}{81} (C_\tau^\Phi - C_\tau) + \frac{a^4 t^3}{1458} S_\tau + O(t^{7/2}), \quad \log W_\tau^{(1)} = B_1 \left[-\frac{a^2 t^3}{1296 \tau^2} + \frac{a^4 t^3}{5832 \tau} \right] \quad (3.7)$$

with the free flowed propagator $(C_\tau = \langle N^{-1} \text{tr} F_{12}^2 \rangle_\tau)$, $C_\tau^\Phi = \langle N^{-1} \text{tr}(e \cdot D\Phi_{n_0})^2 \rangle_\tau$, $S_\tau = \langle N^{-1} \text{tr} \Phi_{n_0}^2 \rangle_\tau$, and an independent expansion of the flowed one-loop Maldacena–Wilson integral for $\tau \gg t$ reproduces it term by term through the exact identities $M_0 = (L_{\text{trace}} - L_{\text{chord}})^2$ and $M_2 = 2\mathcal{A}^2$. The agreement is not circular: one side is the exact first equation, the other a Feynman integral.

3.5 Continuum one loop and the flow regimes

With cusp neighbourhoods of size ε excised,

$$\log W = B_1 \left[-A(s) \log \frac{\ell}{\varepsilon} + F(s) \right] + O(\varepsilon), \quad (3.8)$$

$$A(s) = \sum_{\text{cusps}} 2(\pi - \alpha_c) \tan \frac{\alpha_c}{2} = \frac{2\pi}{3}s + A_2 s^2 + \dots, \quad F(s) = \frac{4\pi}{3}s + O(s^2 \log \frac{1}{s}),$$

$A_2 \approx -0.224$. The exactness of A is tested without fitting: after subtracting it, the successive ε -differences halve. So $W \rightarrow 1$ like $1 - (2\pi B_1/3)|s|(\log(\ell/\varepsilon) - 2)$, non-analytic in a because $a \rightarrow -a$ is a symmetry. With a gradient-flow regulator of resolution τ there are three regimes: regime I ($\tau \gg t$) is the $a^2 t^3$ law above; regime II ($s^2 t \ll \tau \ll t$) gives $\log W \approx -B_1(\sqrt{2\pi}/27)s^2 \sqrt{t/\tau}$; regime III ($\tau \ll s^2 t$) gives $\log W \approx -B_1(2\pi s/3)[\log(8\kappa) + \gamma_E/2 - 2]$ with $\kappa = s\sqrt{t}/(6\sqrt{8\tau})$. Separation-based regulators cut the near-BPS cusp interaction where the sliver width reaches $\sqrt{\tau}$, at distance $\approx \sqrt{\tau}/\alpha$ from the vertex, which produces an $|s| \log |s|$ term absent from vertex excision; the scheme-independent content at $O(s)$ is $A(s)$. Every regime is regular, and the order of the limits $t \rightarrow 0$ and $\tau \rightarrow 0$ changes the exponent but never produces a divergence.

3.6 Rounded family and closure

Replacing each cusp by a circular fillet of radius δ , with the internal vector turning in step with the tangent, gives a C^1 loop with a finite one-loop value

$$\log W^{(1)} = -B_1 \left[A(s) \log \frac{1}{\hat{\delta}} + \frac{2\pi|s|}{3} \left(\log \frac{|s|}{3} - 2 \right) - h|s| \right] + O(s^2), \quad \hat{\delta} = \delta/\sqrt{t}, \quad h = -4.0202, \quad (3.9)$$

valid for $\hat{\delta} \ll |s|$. The coefficient of $\log(1/\hat{\delta})$ is the exact cusp sum and the $|s| \log |s|$ term has the same origin as in regime III; the constant h is convention dependent (an independent implementation gave -4.02025).

The second equation is not closed. At a finite regulator the Schwinger–Dyson identities reduce only the curvature-current part ($\dot{q}^1 E_2 - \dot{q}^2 E_1$), the scalar Laplacian and the radial derivatives, leaving the transverse gauge currents, the transverse scalar Laplacian $(D_3^2 + D_4^2)\Phi$, scalar and fermion currents, scalar-potential and Yukawa insertions, scalar commutators from transport along the chord, the tip gradient and the ordered pairs. At one loop the field-equation part of a derivative moment is a pure regulator contact term while the transverse part has a long-range $4/r^4$ kernel, so the residual cannot be dropped even perturbatively. The flipped hierarchy is an honest infinite hierarchy with a specific residual inventory.

Ledger

Statement	Status
Conformal collapse, Proposition 3.1	Proof given conformal invariance
Constant-scalar completion diverges as $1/s$	One loop: labelled computation; all orders: proof sketch
Zarembo family trivial	Literature
Identity at $a = 0$; evenness in a	Proof
Propositions 3.2, 3.3	Proof (smooth fields); numerically verified
Short-time formula and one-loop confirmation	Proof sketch (remainder not bounded); independent one-loop check
Continuum $A(s)$; $F(s)$ leading term	Proof (cusp formula); exactness tested; $s^2 \log$ term an observation
Regimes I–III	Analytic (thin sliver, moments) plus numerics
Rounded family; h	Numerical verification; h a labelled constant
Closure inventory	Structural reading with one-loop kernel controls

4 The near-BPS corner of the generalized cusp

The continuum short-time law of the flipped loop is $-\sum_c \Gamma(\pi - \alpha_c, \pi) \log(\ell\mu)$ with $\alpha_c \approx |s|/3$, so its coupling dependence is that of the generalized cusp anomalous dimension $\Gamma(\varphi, \theta)$ in the corner $\varphi = \pi - u$, $\theta = \pi - xu$, $u \rightarrow 0$. Two exact inputs control it.

4.1 Exact slope at the BPS line

At every loop order Γ is a polynomial in $\cos \theta$ vanishing on the BPS line $\cos \theta = \cos \varphi$: $\Gamma = \sum_{k \geq 1} (\cos \theta - \cos \varphi)^k R_k(\varphi)$. The CHMS near-BPS formula [1], derived from the latitude matrix model, determines R_1 exactly: $R_1(\varphi) = -2\varphi B(\tilde{\lambda}) / ((1 - \varphi^2/\pi^2) \sin \varphi)$ with $\tilde{\lambda} = \lambda(1 - \varphi^2/\pi^2)$. In corner variables $\cos \theta - \cos \varphi \approx -(1 - x^2)u^2/2$, which must be written as $-2 \sin \frac{\theta + \varphi}{2} \sin \frac{\theta - \varphi}{2}$ in any numerical evaluation, $\sin \varphi \approx u$ and $\tilde{\lambda} \approx 2\lambda u/\pi$, so

$$\Gamma_1 = (1 - x^2) \frac{\pi^2}{2} B\left(\frac{2\lambda u}{\pi}\right) (1 + O(u)) : \quad (4.1)$$

the slope at the BPS line, $-\partial_x \Gamma|_{x=1} = \pi^2 B(2\lambda u/\pi)$, is exact at all couplings and depends on the single effective coupling $g = \lambda u$.

4.2 Two loops and all orders

The planar two-loop cusp of Drukker and Forini [7], transcribed and validated by three independent limits (small angles, the BPS-line slope at every φ , the antiparallel logarithm $\lambda^2/8\pi^3$), gives in the corner

$$\Gamma(\pi - u, \pi - xu) = (1 - x^2) \left[\frac{g}{16\pi} - \frac{g^2}{192\pi^2} \right] + O(gu) + O(g^2 u \log u). \quad (4.2)$$

Through $O(g^2)$ this equals $(1 - x^2)(\pi^2/2)B(2g/\pi)$. Hence $H := \lim_{u \rightarrow 0} \Gamma(\pi - u, \pi)/u = \lambda(1 - 1/N^2)/16\pi$ has no two-loop correction, at finite N as well since the two-loop cusp is pure $C_F C_A$.

Proposition 4.1 (H to all orders in perturbation theory). *Write $\xi = (\cos \theta - \cos \varphi)/\sin \varphi$ and $\Gamma^{(L)} = \sum_k \xi^k c_k^{(L)}(\varphi)$. Suppose that at L loops the antiparallel singularities $\Gamma^{(L)}(\pi - u, \theta_j) =$*

$O(u^{-1} \text{polylog } u)$ hold at $L - 1$ distinct fixed θ_j (they hold at one and two loops and in the ladder limit). Then $c_k^{(L)}(\pi - u) = O(u^{k-1} \text{polylog } u)$ for $k \geq 2$, and $H(\lambda, N) = \lambda(1 - 1/N^2)/16\pi$ order by order in perturbation theory.

The bound follows by inverting a Vandermonde matrix; only $o(u^{-3})$ is needed. The statement is order by order: it should not be called exact nonperturbatively, and the two-loop form $(1 - x^2)(\pi^2/2)B(2g/\pi)$ should not be extrapolated beyond two loops.

4.3 Strong coupling: the classical corner

For the planar classical string with ansatz $z = r\zeta(\hat{\varphi})$ and an S^5 angle, the opening $\Omega = \pi - \varphi$, the internal angle θ and $\Gamma/\sqrt{\lambda}$ are one-dimensional integrals over (ζ_0, r_1) ; the antiparallel limit gives $\Gamma\Omega \rightarrow -4\pi^2/\Gamma(\frac{1}{4})^4$, the small-angle limit $B = \sqrt{\lambda}/4\pi^2$, and the curve $r_1 = 1/\zeta_0^2$ is the BPS line $\theta = \pi/\sqrt{1 + \zeta_0^2} = \pi - \Omega$ with $\Gamma \equiv 0$. In the corner the string is not thin: its depth is $\zeta_0 \sim \sqrt{\Omega}$ while the opening is Ω . With $\zeta_0 = \sqrt{w\Omega}$,

$$x = \frac{3}{\pi w} - \frac{\pi w}{4}, \quad \Gamma = \sqrt{\lambda\Omega} F, \quad F = \frac{1}{4} \left(\sqrt{w} - \frac{4}{\pi^2 w^{3/2}} \right), \quad (4.3)$$

valid for $2\lambda\Omega/\pi \gg 1$, $\Omega \ll 1$. F vanishes at the BPS point $w = 2/\pi$; $F(0) = 1/(\sqrt{2\pi} 3^{3/4}) = 0.17501$; and $F'(1) = -\frac{1}{4}\sqrt{2/\pi}$, which equals $-\pi^2 B(2g/\pi)$ at large g : the classical corner reproduces the one exactly known coefficient. F is not proportional to $1 - x^2$ (the $k = 1$ part alone would give $F(0) \approx 0.0997$). The physical branch $F(|x|)$ has a classical kink at $\theta = \pi$, where antipodal points of S^5 are joined by an S^4 of great circles; quantum effects should smooth it. A byproduct is the antiparallel-lines law near $\theta = \pi$ at strong coupling, $V_{\parallel} \approx -\sqrt{\lambda}(\pi - \theta)^{3/2}/(3^{3/2}\sqrt{\pi}L)$, non-analytic in $\cos\theta$; we did not find it in the literature.

4.4 Consequences for the flipped loop

Per cusp $\Gamma_c = \Phi(\lambda\alpha_c, x = 0)$ with $\alpha_c \approx |s|/3$. For $\lambda|s| \ll 1$ the continuum law $\log W \approx -(\lambda(1 - 1/N^2)|s|/24\pi) \log(\ell\mu)$ is one-loop exact perturbatively; for $1/\lambda \ll |s| \ll 1$ (planar) $\log W \approx -0.2021\sqrt{\lambda|s|} \log(\ell\mu)$. Since $s = a\sqrt{t}$, the Loewner time sets the effective coupling of the corner, $g(t) = \lambda|a|\sqrt{t}/3$: in a strongly coupled theory the continuum short-time law changes from $|s| \log$ to $\sqrt{|s|} \log$ as t grows past $\sim 9/(\lambda a)^2$. The crossover function $\Phi(g, 0)$ is not determined beyond $O(g^2)$; the three-loop generalized cusp [2] would fix the next term.

Ledger

Statement	Status
Exact $k = 1$ part and BPS-line slope	Proof given CHMS eq. 8 (literature)
Two-loop corner; H without two-loop term	Proof from the DF formulas, validated by three limits
H one-loop exact to all orders	Order-by-order argument under the multi-angle bound (assumption)
Classical corner $F(x)$, $F(0)$, $F'(1)$	Derived (leading order in Ω), numerically checked; $\lambda\Omega \gg 1$
$V_{\parallel} \propto (\pi - \theta)^{3/2}$	Derived; not located in the literature
Loewner time as effective coupling	Corollary

5 A finite-mass source: memory, energy and an exactly local boundary field

Locality about the straight line (Proposition 2.2) is a statement about an infinitely heavy source. A source of finite mass m has an internal scale, and at strong coupling it is described by a string ending on a flavour brane at $z_m = \sqrt{\lambda}/2\pi m$ [8, 9]. We linearize that system exactly.

5.1 Setup and exact endpoint relations

Poincaré AdS₅, $ds^2 = z^{-2}(-dt^2 + d\vec{x}^2 + dz^2)$ with $R = 1$, tension $\sqrt{\lambda}/2\pi$; the string hangs from z_m to the Poincaré horizon, with static energy $m = (\sqrt{\lambda}/2\pi) \int_{z_m}^{\infty} dz/z^2$. An external force acts through $S_F = \int dt F(t)x(t, z_m)$, equivalently the boundary condition $\Pi_x^z|_{z_m} = F$ of [8]. One transverse direction is kept. The quadratic action is $S_2 = -(\sqrt{\lambda}/4\pi) \int dt dz z^{-2}(x'^2 - \dot{x}^2)$, the bulk equation is $\ddot{x} = x'' - (2/z)x'$, and the boundary variation gives $x'(t, z_m) = -(z_m/m)F(t)$ (a positive force makes the string trail). The mode equation is Bessel of order $3/2$, hence elementary, and the retarded branch is

$$x(t, z) = \tilde{X}(t - z) + z\tilde{X}'(t - z), \quad (5.1)$$

the linearization of Mikhailov's solution [3]; the other branch is a wave emerging from the horizon, excluded by the vacuum initial condition. Let $Y(t) := \tilde{X}(t - z_m)$, the retarded auxiliary trajectory.

Proposition 5.1 (exact linearized dynamics). *With $X(t) = x(t, z_m)$ the physical endpoint position,*

$$F = m\ddot{Y}, \quad X = Y + z_m\dot{Y}, \quad m\ddot{X} = F + z_m\dot{F}. \quad (5.2)$$

The third relation is the nonrelativistic, small-force limit of the Chernicoff–García–Güijosa equation (their eq. 28 of [9], whose linearization we recover independently); their dispersion relation gives $p_q = mv - z_m F = m\dot{Y}$, so the auxiliary velocity is the momentum of the dressed quark. The auxiliary trajectory is the causal exponential smearing of the physical one, $Y = (1 + z_m\partial_t)^{-1}X$ with kernel $z_m^{-1}e^{-s/z_m}\theta(s)$, so the force needed to sustain a prescribed trajectory has the susceptibility

$$\chi_m(\omega) = \frac{-m\omega^2}{1 - i\omega z_m} = -m\omega^2 - i\frac{\sqrt{\lambda}}{2\pi}\omega^3 + \frac{\sqrt{\lambda}}{2\pi}z_m\omega^4 + \dots, \quad (5.3)$$

with a single pole at $\omega = -i/z_m$. Its first two terms are the mass and Abraham–Lorentz terms of (2.2) at $m_R = m$, $2\pi B = \sqrt{\lambda}/2\pi$; the rest is the memory, on the Compton scale z_m of the dressed quark. At $\omega z_m \gg 1$ the endpoint is a pure damper, $|\chi_m| \rightarrow (2\pi m^2/\sqrt{\lambda})\omega$. Rationality holds for endpoint quantities at $T = 0$, in the strict planar limit (rigid flavour brane), at strict $\lambda \rightarrow \infty$ and at linear order.

5.2 Energy account

Proposition 5.2 (positive finite-time account). *With $\dot{Y}(-\infty) = 0$,*

$$W(T) = \int_{-\infty}^T F\dot{X} dt = \frac{m}{2}\dot{Y}(T)^2 + \frac{\sqrt{\lambda}}{2\pi} \int_{-\infty}^T \ddot{Y}^2 dt \geq 0 \quad \text{for every } T, \quad (5.4)$$

and $(\sqrt{\lambda}/2\pi)\ddot{Y}^2 = (\sqrt{\lambda}/2\pi m^2)F^2$ is the CGG radiation rate. Both terms are the string's own energy: $\int_{z_m}^{\infty} \mathcal{H} dz = W(T)$ exactly, the first term being a total z -derivative of the energy density and the second the energy of the wave in flight. As $m \rightarrow \infty$ at fixed λ ,

$$\frac{m}{2}\dot{Y}^2 = \frac{m}{2}\dot{X}^2 - \frac{\sqrt{\lambda}}{2\pi}\dot{X}\ddot{X} + \frac{\sqrt{\lambda}}{4\pi}z_m(\ddot{X}^2 + 2\dot{X}\ddot{X}) + O(\sqrt{\lambda}z_m^2). \quad (5.5)$$

The Schott energy of Proposition 2.4 is the $O(z_m)$ truncation of a perfect square: finite-time non-passivity of the point limit is an artefact of the truncation. For the drive $(1 - s^2)^5$ the renormalized point account is negative just after onset while (5.4) is nonnegative (records, group D).

5.3 The boundary field is exactly local in the retarded time

The gauge-theory field of the moving source is read at the boundary through bulk-to-boundary propagators. For the dilaton-coupled operator O ($\Delta = 4$) the Euclidean propagator is $K_E = Cz^4/(z^2 + \rho^2 + \tau_E^2)^4$ and its retarded version, with the Kubo sign, is

$$K_R(\tau, \rho, z) = -\frac{\pi}{3}C\theta(\tau)z^4\delta'''(z^2 + \rho^2 - \tau^2), \quad (5.6)$$

normalized so that $\int d\tau K_R = \int d\tau_E K_E$; it is supported on the light cone only, because $(s + i\varepsilon)^{-4}$ is real for $s \neq 0$, a property of the integer dimension. The string sources O with density z^{-2} at transverse position $x(t', z)\hat{e}$, so to first order

$$\langle O \rangle^{(1)}(t, \vec{r}) = \frac{2\pi}{3}C(\vec{r} \cdot \hat{e}) \int_{z_m}^{\infty} dz z^2 \partial_s^4 \left[\frac{x(t - \sqrt{s}, z)}{2\sqrt{s}} \right]_{s=z^2+r^2}, \quad (5.7)$$

which for a constant displacement reproduces the gradient of the static profile $C(5\pi/16) \int dz z^2 (z^2 + r^2)^{-7/2}$ (Coulombic for $r \gg z_m$, regular at $r \rightarrow 0$: the gluonic cloud of [9]). For $Y = e^{-i\omega t}$, with $g_4(s; \omega) = \partial_s^4 [e^{i\omega\sqrt{s}}/2\sqrt{s}] = e^{i\omega\rho}\rho^{-9}[(\omega\rho)^4 + 10i(\omega\rho)^3 - 45(\omega\rho)^2 - 105i\omega\rho + 105]/32$, the transfer function is $\mathcal{T}(\omega; r, z_m) = \int_{z_m}^{\infty} dz z^2 (1 - i\omega z) e^{i\omega(z-z_m)} g_4(z^2 + r^2; \omega)$. The delays $u(z) = z - z_m + \sqrt{z^2 + r^2}$ increase from $r_m = \sqrt{r^2 + z_m^2}$, so a continuum of delays appears to contribute.

Proposition 5.3 (exact localization). *The integrand of $\mathcal{T}$ is an exact z -derivative, dG/dz , with $\rho = \sqrt{z^2 + r^2}$,*

$$G(z) = -e^{i\omega(z+\rho)} \frac{N(z, \rho, \omega)}{32\rho^7(\rho+z)^3}, \quad (5.8)$$

$$N = \omega^4 \rho^3 z^3 (\rho+z)^2 + 2i\omega^3 \rho^2 z^2 (\rho+z)^2 (2\rho+3z) - \omega^2 \rho z (\rho+z) (8\rho^3 + 24\rho^2 z + 33\rho z^2 + 15z^3) - i\omega(\rho+z)S_4 + S_4, \quad (5.9)$$

$S_4 = 8\rho^4 + 24\rho^3 z + 48\rho^2 z^2 + 45\rho z^3 + 15z^4$, and $G \rightarrow 0$ as $z \rightarrow \infty$. Hence

$$\mathcal{T}(\omega; r, z_m) = e^{i\omega r_m} P_4(\omega), \quad P_4(\omega) = \frac{N(z_m, r_m, \omega)}{32r_m^7(r_m + z_m)^3}, \quad (5.10)$$

a quartic polynomial times a single delay: the boundary response is a fourth-order differential operator on Y at the bulk null travel time r_m from the endpoint to the boundary point.

G was found by a polynomial ansatz with $(\rho^2 - z^2)$ denominators and verified by symbolic differentiation and, numerically, against contour-rotated quadrature and a direct time-domain pulse computation (records, group G). For $z_m \rightarrow 0$, $P_4 \rightarrow (1 - i\omega r)/(4r^6)$: the boundary point at distance r reads $Y(t - r) + r\dot{Y}(t - r) = x(t, z = r)$, the string at height r at the same time. The static value $P_4(0) = S_4/(32r_m^7(r_m + z_m)^3)$ is a closed form for $(105/32) \int_{z_m}^{\infty} z^2 (z^2 + r^2)^{-9/2} dz$. In terms of the physical trajectory $\hat{\mathcal{T}}_X = e^{i\omega r_m} P_4(\omega)/(1 - i\omega z_m)$, and since N is not divisible by $(1 - i\omega z)$ the exponential memory is present and is the only memory. The mechanism is that the propagator is light-cone supported and the retarded string $x = Y(u) + z\dot{Y}(u) = -z^2 \partial_z [Y(u)/z]$ is a congruence of null rays from the endpoint, whose imprints telescope; this is the dilaton analogue of the light-cone localization of the energy density of the Mikhailov string [10, 11, 12]. Whether it persists at nonlinear order is open.

Remark 5.4. A referee context reports that Callan and Güijosa [13] computed this observable at $z_m = 0$, obtained an intermediate kernel that agrees with $32g_4e^{-i\omega\rho}$ only at $\omega = 0$, and concluded an infinitely broadened pulse. If that reading, made through a text extractor, is confirmed by eye against the journal version, Proposition 5.3 corrects it. We make no stronger statement here.

5.4 Verdict for the front test

At linear order the finite-mass channel is a rational transfer with one simple pole, times one fixed light-travel delay; the exponents are integers fixed by the wave equation and by the integer dimension of O , and the single scale z_m rescales time without changing the shape ($\chi_m z_m^3/\sqrt{\lambda}$ depends on ωz_m alone). There is no power front, no logarithmic tangent and no hierarchy of delays. The channel fails the front-and-tangent test before any prime is involved, and no arithmetic comparison is run. A finite temperature would add a quasinormal tower of poles $\omega_n \propto T$, not logarithms of primes.

Ledger

Statement	Status
Proposition 5.1	Proof, exact at linear order; referee confirmed
Agreement with CGG eqs. 28, 34, 35 in the nonrelativistic limit	Check of transcribed formulas; equation numbers to confirm by eye
χ_m rational, one pole; expansion	Proof
Proposition 5.2; string-energy identity; Schott truncation	Proof (asymptotic series for the truncation)
Retarded $\Delta = 4$ propagator	Proof, confirmed two ways
Proposition 5.3; massless limit; static closed form	Proof (symbolic differentiation), numerically verified
Callan–Güijosa discrepancy	Referee reading; unverified by eye
Front test failure	Proof, under the stated qualifications
Nonlinear localization	Open

6 The quartic straight-line response

The first nonlocal response about the straight line is quartic in the sources (Proposition 2.1 has no cubic term). We analyse its structure, validate the strong-coupling input, and state what it can and cannot provide.

6.1 Operator content of the tilt channel

Deform the internal direction to $n(\sigma) = (j_a(\sigma), \sqrt{1-j^2})$ on S^5 , $a = 1, \dots, 5$, at $|\dot{x}| = 1$. The coupling is $\exp \int [j_a \Phi_a + (\sqrt{1-j^2} - 1)\Phi_6]$, and odd orders vanish by $SO(5)$. The $O(j^4)$ term of $\log Z$ has four pieces:

$$\frac{1}{4!} \int^4 j j j j \langle \Phi_a \Phi_b \Phi_c \Phi_d \rangle_{\text{conn}} - \frac{1}{4} \int^3 j_a j_b j^2 \langle \Phi_a \Phi_b \Phi_6 \rangle + \frac{1}{8} \int^2 j^2 j^2 \langle \Phi_6 \Phi_6 \rangle_{\text{conn}} - \frac{1}{8} \int j^4 \langle \Phi_6 \rangle. \quad (6.1)$$

Φ_6 inserted on the half-BPS line is the lowest non-protected operator [14, 15], with

$$\Delta_6 = 1 + \frac{\lambda}{4\pi^2} + O(\lambda^2), \quad \Delta_6 = 2 - \frac{5}{\sqrt{\lambda}} + \frac{295}{24\lambda} - \frac{305}{16\lambda^{3/2}} + \left(\frac{351845}{13824} - \frac{75}{2}\zeta(3) \right) \frac{1}{\lambda^2} + \dots \quad (6.2)$$

at weak and strong coupling respectively [15, 16, 18, 19], and at strong coupling $\Phi_6 \leftrightarrow y^a y^a$. Remarks: $\langle \Phi_6 \rangle$ vanishes in a conformal scheme but is a linear divergence in a cutoff scheme, needed to make the

tree-level quartic term finite; the Φ_6 terms are contact-type (two source times coincide), and at weak coupling $\langle\langle\Phi_6\Phi_6\rangle\rangle$ is the leading quartic term, $O(\lambda)$, giving $j_a(t) \int dt' j(t')^2 |t-t'|^{-2\Delta_6}$; the connected four-point function is $O(\lambda^2)$ at weak and $O(1/\sqrt{\lambda})$ at strong coupling in unit normalization, to be multiplied by $(2B)^2$.

6.2 Structural propositions

Proposition 6.1 (the generalized free field is linear). *In the $\lambda \rightarrow \infty$ limit the defect fields are generalized free, $[\Phi(t), \Phi(0)]$ is a c -number, and every nested commutator vanishes. The cubic retarded response at separated source times is therefore entirely the connected four-point function, $O(1/\sqrt{\lambda})$: equivalently, $\log Z = -S_{\text{cl}}[n] + O(\lambda^0)$ and its quartic term is the tree Witten diagram of the sphere fluctuations on AdS_2 .*

Proposition 6.2 (memory). *The connected function at $O(1/\sqrt{\lambda})$ contains $\log|\chi|$ and $\log|1-\chi|$ with nonzero coefficients ($-2/5, -1, +1$ at $\chi \rightarrow 1$ in the singlet, symmetric and antisymmetric channels), whereas the generalized-free part is rational. The Lorentzian continuation of a rational function gives contact-supported commutators; that of the logarithms gives step-function discontinuities. The cubic response therefore has memory at $O(1/\sqrt{\lambda})$, and since the $\log \chi$ terms are $\gamma \log \chi$ from $\chi^{\Delta_0 + \gamma/\sqrt{\lambda}}$, that memory is tied to the anomalous dimensions, which resum to power laws with coupling-dependent exponents.*

Proposition 6.3 (no scale). *Every retarded response kernel about the straight line is homogeneous in the time differences (the cubic tilt kernel has degree -4); anomalous-dimension logarithms appear as $\log(t_{ij}/t_{kl})$; a constant background tilt is an $SO(6)$ rotation of the half-BPS line. Hence no fixed delay, and in particular no hierarchy at $\log 2, \log 3, \dots$, can arise about the straight line at any order. A time-dependent background, a curved contour, a temperature or a smeared observable would introduce a scale.*

The Lorentzian cubic kernel itself, which needs all Wightman orderings (the continuation across $\chi = 0, 1, \infty$ with $i\epsilon$ prescriptions), was not computed.

6.3 Validation of the strong-coupling input

The tree-level connected four-point functions of the $\Delta = 1$ fields y^a (the superprimary of the displacement multiplet) and of the $\Delta = 2$ fields x^i were taken from [16] through an automated fetch summary and were tested rather than trusted. With $\chi = t_{12}t_{34}/(t_{13}t_{24})$, $G = G^{(0)} + G^{(1)}/\sqrt{\lambda}$ and the S, T, A structures, crossing $1 \leftrightarrow 2$ requires $G_{S,T}(\chi) = G_{S,T}(\chi/(\chi-1))$, $G_A(\chi) = -G_A(\chi/(\chi-1))$, and crossing $1 \leftrightarrow 3$ requires $G(\chi) = (\chi/(1-\chi))^{2\Delta} R G(1-\chi)$ with $R_5 = \begin{bmatrix} 1/5 & 28/25 & -4/5 \\ 1/2 & 3/10 & 1/2 \\ -1/2 & 7/10 & 1/2 \end{bmatrix}$ for $SO(5)$

and $R_3 = \begin{bmatrix} 1/3 & 10/9 & -2/3 \\ 1/2 & 1/6 & 1/2 \\ -1/2 & 5/6 & 1/2 \end{bmatrix}$ for $SO(3)$, both with $R^2 = 1$.

$\Delta = 1$. The transcribed functions satisfy both crossings at both orders to 10^{-10} ; $G_S^{(1)} = -2\chi^2 \log \chi - \frac{43}{30}\chi^2 + O(\chi^3 \log \chi)$ with no constant and no linear term, so the identity is not corrected; and the $\chi^n \log \chi$ coefficients divided by the generalized-free OPE coefficients $2/5, 3/5, -1$ give $\gamma_S = -5$ (the strong-coupling Δ_6), $\gamma_T(\Delta_0 = 4) = -5$ and $\gamma_A(\Delta_0 = 3) = -4$, with no $\chi^2 \log \chi$ in the symmetric channel as the protected $\Delta = 2$ operator requires. The transcription is validated. Peeling off blocks order by order, a referee context found $\gamma_S(n) = -(2n^2 + 3n + 5)$, $\gamma_T(n) = -(2n^2 + 3n)$, $\gamma_A(n) = -(2n^2 + 5n + 4)$, to be compared with [16].

$\Delta = 2$. The generalized-free part passes; the transcribed tree functions violate 16 of 18 crossing relations and give $\gamma_T(\Delta_0 = 4) = -40/9$ instead of -5 . That transcription is unreliable and is

not used. The displacement four-point function is in any case fixed by the superconformal Ward identities from the $\Delta = 1$ function [17, 18].

For all-equal internal indices $G_{1111} = G_S + \frac{8}{5}G_T$ and $G_{1111}^{(1)} = -2\chi^2 \log \chi + \dots$, so the quartic tilt integrand $G^{(1)}/(t_{12}^2 t_{34}^2)$ is integrable at coincident points.

6.4 Verdict

The quartic response is the first nonlocal one on the straight line and its memory exponent, set by $\Delta_6(\lambda) \in (1, 2)$, is coupling dependent: a fixed protected dimension gives no variable exponent, but a non-protected one does. However, the response is nonlinear, and the ordered tests presuppose a linear channel; a linear channel would need a background supplying a scale; by Proposition 6.3 no channel about the straight line has fixed delays, so the prime-delay test cannot be passed in principle; and a dictionary $\lambda \leftrightarrow \omega$ would be an adjustment of the theory to the target. No arithmetic comparison is run.

Ledger

Statement	Status
Operator content of the quartic tilt response	Proof (expansion); referee confirmed
$\Delta_6(\lambda)$ expansions	Literature, transcribed
Proposition 6.1	Proof
Proposition 6.2	Proof sketch (discontinuity structure); kernel not computed
Proposition 6.3	Proof (dilatations)
Crossing matrices	Proof; referee confirmed by index sums
$\Delta = 1$ transcription validated	Numerical verification (93 cases)
$\Delta = 2$ transcription unreliable	Numerical falsification
Anomalous-dimension patterns	Referee observation; compare with [16]

7 Assessment and outlook

7.1 What memory exists in the half-BPS line

Three kinds of memory were met. The Compton-scale exponential of a massive source (Section 5) is rational: one pole, one fixed light-travel delay. The anomalous-dimension power laws of non-protected operators (Section 6) are scale-free and enter first at cubic order in the sources. A finite temperature would add a quasinormal tower, again rational poles. None has a power front with a variable exponent accompanied by a logarithmic tangent, and none has a hierarchy of fixed delays. The growing-trace observables of Sections 3–4 are Euclidean expectation values on a one-parameter shape family with no input/output structure at all.

7.2 The arithmetic reading

Against the ordered tests of Section 1: the linear straight-line channel fails at the front (it is a polynomial); the finite-mass channel fails at the front (it is rational); the quartic channel is not linear and has no scale, so it cannot pass the delay test even in principle; the Loewner family is not a channel. The one structural rhyme with the arithmetic side is that passivity holds cumulatively but not instantaneously in the point limit (Proposition 2.4), and Section 5 shows that even this is an artefact of the infinite-mass truncation. We therefore run no arithmetic comparison and regard the

displacement and Loewner channels of the half-BPS line as closed for that purpose, at the orders and in the backgrounds examined. The closure does not extend to backgrounds with a derived scale (a rotating internal direction, a cusp, a temperature), where a linear channel with memory could be constructed; any such construction must derive its scale rather than choose it, and would still face the absence of any mechanism for logarithms of primes in a conformal defect.

7.3 The physics that stands

Independently of the arithmetic goal, the following are results: the locality theorem for the linear response of any conformal line defect with a dimension-two displacement; the exact evolution equations and regular short-time laws of the flipped-return Loewner loop, with the Loewner time acting as the effective coupling of a near-BPS corner; the one-loop exactness of the near-BPS corner coefficient order by order, its $\sqrt{\lambda u}$ strong-coupling window with an explicit classical function that reproduces the exactly known BPS-line slope, and the $(\pi - \theta)^{3/2}$ antiparallel law; the exact linearized dynamics and positive energy account of the finite-mass source, exhibiting the Schott term as a truncation; and the exact light-cone localization of the dilaton-coupled boundary field with its closed-form quartic transfer function.

7.4 Open items

1. Confirm the Callan–Güijosa comparison by eye, and test whether the dilaton localization persists for the nonlinear Mikhailov string.
2. Compute the Lorentzian cubic kernel of the tilt channel from the validated $\Delta = 1$ four-point function, or from the Lorentzian tree diagram on AdS_2 , to make Proposition 6.2 quantitative.
3. Extract the three-loop corner $\Phi(g, 0)$ from the three-loop generalized cusp [2] and the one-loop string correction that smooths the kink of F at $\theta = \pi$.
4. Obtain an independent specialist review of the whole; every audit so far was by the same assistant.

A Numerical evidence, provenance and acknowledgements

A.1 Programs and records

Seven standard-library Python programs accompany the text, each printing one JSON object, deterministic (byte-identical on repeated runs on a given platform), with records under `N4SYM/numerics/records/`. Every case is a floating diagnostic of an analytic statement; the $SU(2)$ backgrounds are classical test connections, not samples of the quantum measure; no zeta function is evaluated anywhere.

Program	Record	Cases	Section
<code>check_straight_line_response.py</code>	<code>straight-line-response-20260923.json</code>	72	2
<code>check_trace_completion.py</code>	<code>trace-completion-20260923.json</code>	77	3
<code>check_flipped_return.py</code>	<code>flipped-return-20260924.json</code>	47	3
<code>check_near_bps_corner.py</code>	<code>near-bps-corner-20260924.json</code>	56	4
<code>check_rounded_and_closure.py</code>	<code>rounded-and-closure-20260924.json</code>	20	3
<code>check_finite_mass_response.py</code>	<code>finite-mass-response-20260924.json</code>	107	5
<code>check_quartic_input.py</code>	<code>quartic-input-20260924.json</code>	93	6
Total		472	

All 472 cases pass. The programs are not yet registered in a validation replay; registration belongs to the pass that records a reviewed build. Traps recorded in the research notes include: the implicit linear-driver relation is ill-conditioned for $s < 0.05$ (use the exact series); $\cos \theta - \cos \varphi$ must be written as $-2 \sin \frac{\theta+\varphi}{2} \sin \frac{\theta-\varphi}{2}$ near the corner; the classical corner string is not thin; fillet rounding and flow are separation-based regulators and give $|s| \log |s|$, vertex excision does not; the Zarembo-stadium control has an identically vanishing integrand; the oscillatory depth integral of Section 5 should be evaluated on a rotated contour, or through its antiderivative.

A.2 Provenance of literature inputs

Publication data were taken from the INSPIRE API. The CHMS normalizations and near-BPS formula, the Drukker–Forini two-loop cusp, the Chernicoff–García–Güijosa equations, the Giombi–Roiban–Tseytlin four-point functions and the Beccaria–Giombi–Tseytlin and Ferrero–Meneghelli dimensions were read through automated fetch summaries of the arXiv versions. Each formula used in a derivation was validated before use: the DF cusp by three limits, the CGG equations by their nonrelativistic limits against an independent derivation, the GRT $\Delta = 1$ functions by crossing in two channels and by the OPE, the GRT $\Delta = 2$ functions likewise (and rejected). Equation numbers cited from these summaries should be confirmed by eye before publication.

A.3 Research record

The research notes, with statement ledgers, the same-assistant audits with separate referee contexts, and the continuation notes are in `N4SYM/notes` and `N4SYM/reviews`. The referee passes found and corrected seven, six and three items in the three sessions respectively; the corrections are incorporated here.

A.4 Acknowledgements and review status

This manuscript was drafted for Edward Baker with substantial LLM assistance: the first two research sessions and their notes were prepared with Claude Opus 5.5 (Anthropic) and the third session and this draft with Claude Fable 5.1 (Anthropic; session configuration `claude-fable-5-1`, the serving model can differ), including derivations, diagnostic programming, source checking and document preparation; the research proposal that opened the investigation was prepared with GPT-6 (Codex). Reasoning-effort settings were not exposed and are not inferred. The audits were performed by the same assistants with separate referee contexts; no independent specialist review is claimed. Established results are attributed to their sources, and the assumptions behind every order-by-order or proof-sketch statement are stated in the ledgers.

References

- [1] D. Correa, J. Henn, J. Maldacena and A. Sever, *An exact formula for the radiation of a moving quark in $N=4$ super Yang Mills*, JHEP **06** (2012) 048, [arXiv:1202.4455](#).
- [2] D. Correa, J. Henn, J. Maldacena and A. Sever, *The cusp anomalous dimension at three loops and beyond*, JHEP **05** (2012) 098, [arXiv:1203.1019](#).
- [3] A. Mikhailov, *Nonlinear waves in AdS/CFT correspondence*, [arXiv:hep-th/0305196](#).
- [4] K. Zarembo, *Supersymmetric Wilson loops*, Nucl. Phys. B **643** (2002) 157–171, [arXiv:hep-th/0205160](#).
- [5] Z. Guralnik and B. Kulik, *Properties of chiral Wilson loops*, JHEP **01** (2004) 065, [arXiv:hep-th/0309118](#).
- [6] A. Dymarsky, S. S. Gubser, Z. Guralnik and J. M. Maldacena, *Calibrated surfaces and supersymmetric Wilson loops*, JHEP **09** (2006) 057, [arXiv:hep-th/0604058](#).
- [7] N. Drukker and V. Forini, *Generalized quark-antiquark potential at weak and strong coupling*, JHEP **06** (2011) 131, [arXiv:1105.5144](#).
- [8] M. Chernicoff, J. A. García and A. Güijosa, *Generalized Lorentz-Dirac equation for a strongly coupled gauge theory*, Phys. Rev. Lett. **102** (2009) 241601, [arXiv:0903.2047](#).
- [9] M. Chernicoff, J. A. García and A. Güijosa, *A tail of a quark in $N=4$ SYM*, JHEP **09** (2009) 080, [arXiv:0906.1592](#).
- [10] C. Athanasiou, P. M. Chesler, H. Liu, D. Nickel and K. Rajagopal, *Synchrotron radiation in strongly coupled conformal field theories*, Phys. Rev. D **81** (2010) 126001; erratum **84** (2011) 069901, [arXiv:1001.3880](#).
- [11] Y. Hatta, E. Iancu, A. H. Mueller and D. N. Triantafyllopoulos, *Radiation by a heavy quark in $N=4$ SYM at strong coupling*, Nucl. Phys. B **850** (2011) 31–52, [arXiv:1102.0232](#).
- [12] V. E. Hubeny, *Holographic dual of collimated radiation*, New J. Phys. **13** (2011) 035006, [arXiv:1012.3561](#).
- [13] C. G. Callan and A. Güijosa, *Undulating strings and gauge theory waves*, Nucl. Phys. B **565** (2000) 157–175, [arXiv:hep-th/9906153](#).
- [14] J. Polchinski and J. Sully, *Wilson loop renormalization group flows*, JHEP **10** (2011) 059, [arXiv:1104.5077](#).
- [15] M. Beccaria, S. Giombi and A. Tseytlin, *Non-supersymmetric Wilson loop in $N=4$ SYM and defect 1d CFT*, JHEP **03** (2018) 131, [arXiv:1712.06874](#).
- [16] S. Giombi, R. Roiban and A. A. Tseytlin, *Half-BPS Wilson loop and AdS_2/CFT_1* , Nucl. Phys. B **922** (2017) 499–527, [arXiv:1706.00756](#).
- [17] P. Liendo, C. Meneghelli and V. Mitev, *Bootstrapping the half-BPS line defect*, JHEP **10** (2018) 077, [arXiv:1806.01862](#).

- [18] P. Ferrero and C. Meneghelli, *Bootstrapping the half-BPS line defect CFT in $N=4$ supersymmetric Yang-Mills theory at strong coupling*, Phys. Rev. D **104** (2021) L081703, [arXiv:2103.10440](#).
- [19] D. Grabner, N. Gromov and J. Julius, *Excited states of one-dimensional defect CFTs from the quantum spectral curve*, JHEP **07** (2020) 042, [arXiv:2001.11039](#).